$$4H_3BO_3 \xrightarrow{373\ K} 4HBO_2 + 4\,H_2O$$
$$4HBO_2 \xrightarrow{413\ K} H_2B_4O_7 + H_2O$$
$$H_2B_4O_7 \xrightarrow{Red\ hot} 2B_2O_3 + H_2O$$

Acton of ammonia

Fusion of urea with $B(OH)_3$, in an atmosphere of ammonia at 800 - 1200 K gives boron nitride.

$$B(OH)_3 + NH_3 \xrightarrow{\Delta} BN + 3H_2O$$

Ethyl Borate test

When boric acid or borate salt is heated with ethyl alcohol in presence of conc. sulphuric acid, an ester, triethylborate is formed. The vapour of this ester burns with a green edged flame and this reaction is used to identify the presence of borate.

$$H_3BO_3 + 3C_2H_5OH \xrightarrow[H_2SO_4]{Conc.} B(OC_2H_5)_3 + 3H_2O$$

Note: The trialkyl borate on reaction with sodium hydride in tetrahydrofuran to form a coordination compound $Na[BH(OR)_3]$, which acts as a powerful reducing agent.

Formation of boron trifluoride:

Boric acid reacts with calcium fluoride in presence of conc. sulphuric acid and gives boron trifluoride.

$$3CaF_2 + 3H_2SO_4 + 2\,B(OH)_3 \longrightarrow 3CaSO_4 + 2BF_3 + 6H_2O$$

Boric acid when heated with soda ash it gives borax

$$Na_2CO_3 + 4B(OH)_3 \longrightarrow Na_2B_4O_7 + CO_2 + 6H_2O$$

Structure of Boric acid:

Boric acid has a two dimensional layered structure. It consists of $[BO_3]^{3-}$ unit and these are linked to each other by hydrogen bonds as shown in the Figure 2.2.

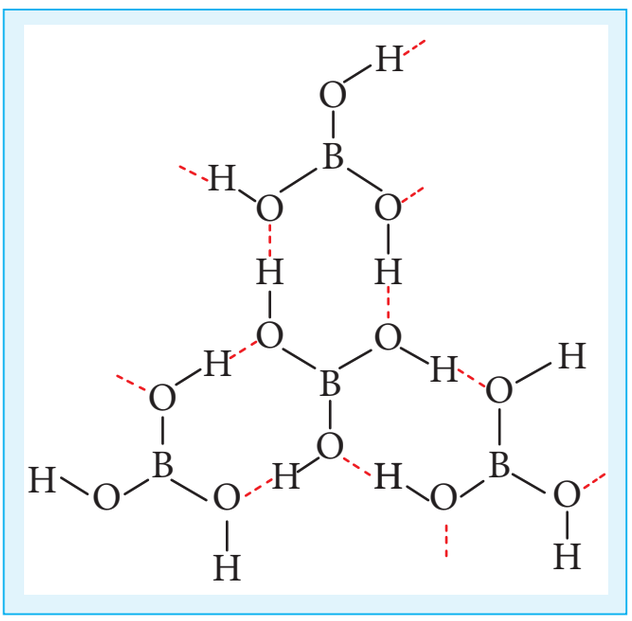


Figure 2.2 Structure of boric acid

Uses of boric acid:

1. Boric acid is used in the manufacture of pottery glases, enamels and pigments.
2. It is used as an antiseptic and as an eye lotion.
3. It is also used as a food preservative.

XII_U2-P-Block.indd 35 04-01-2021 12:48:03